Measures of Disease Occurrence

APHS 20203 • Epidemiology & Biostatistics

Objectives

By the end of this week, students will be able to:

1. Define counts versus rates.
2. Distinguish crude, specific, and adjusted rates.
3. Compute and interpret prevalence and incidence, including cumulative incidence (CI) and incidence rate (IR).
4. Interpret key mortality measures: crude death rate (CDR), cause-specific mortality rate (CSMR), case-fatality rate (CFR), and proportional mortality ratio (PMR).

Measures for Comparison

Counts and Rates

What Does Epidemiology Ask?

Are exposure and outcome linked?

$$\textbf{Exposure} \longleftrightarrow \textbf{Outcome}$$

Open-ended question: Describe a health issue you have heard about recently. What might be the exposure? What is the outcome?

Exposure

1. What is the exposure?
2. Who are the exposed?
3. What are the exposure's potential health effects?
    - Beneficial or harmful?

Outcome

1. Who is getting disease?
2. Where is disease occurring?
3. When is disease occurring?
4. How do we measure disease frequency?
    - What are the potential contributing factors?

Example: Count vs. Rate

Passenger group	Number of passengers	Number of survivors	Survival rate
First-class men		57	
First-class women		140	
First-class children		5	
First-class total		**202**	
Second-class men		14	
Second-class women		80	
Second-class children		24	
Second-class total		**118**	
Third-class men		75	
Third-class women		76	
Third-class children		27	
Third-class total		**178**	
Total		**498**	

Table 1. Likelihood of survival based on class and gender. Raw data from Tokis (1999).

In terms of the physical space occupied by each class of passengers, first- and second-class cabins were much closer to the boat deck, where the lifeboats were located, than third-class cabins were.

Use of Epidemiologic Measures

Counts	Rates
Magnitude: how much	Risk: how likely
Simple tally	Population-based
Planning	Comparison

Discussion: Counts vs. Rates

Think of a real-world public health or health-related example.

- In what situation would using a **count** be more appropriate?
- In what situation would using a **rate** be more appropriate?
- Does a **bigger number** always mean higher **risk**? Why or why not?

Crude, Specific, and Adjusted Rates

- **Crude rates:** The big picture
- **Specific rates:** Zooming in
- **Adjusted rates:** Fair comparisons

Crude Rates: Initial Comparison

What are they?

- Total cases divided by the total population in a given time period

What are they good for?

- Quick, overall comparisons between populations

Key limitation

- Crude rates do not account for population differences such as age.

Crude Rates Example

Age group	Youngtown population	Oldtown population	Youngtown deaths	Oldtown deaths
10-20	20,000	5,000	200	30
20-30	40,000	5,000	150	20
30-40	15,000	10,000	50	40
40-50	10,000	10,000	20	20
50-60	5,000	15,000	15	90
60-70	5,000	25,000	10	150
70-80	3,500	20,000	20	250
80-90	1,500	10,000	35	200
Total	**100,000**	**100,000**	**500**	**800**

$$\text{Crude rate (Youngtown)} = \frac{500}{100{,}000} \times 100{,}000 = 500$$

What stands out about the age distribution in each town?

Discussion

Scenario: Consider two towns with the same population but different death rates in 2022:

- Youngtown: 500 deaths per 100,000 people
- Oldtown: 800 deaths per 100,000 people

Questions:

- Does Oldtown's higher crude death rate mean it is less healthy than Youngtown?
- What population characteristic might explain this difference?

Specific Rates: Zooming into Subgroups

What are they?

- Rates calculated for a specific subgroup, such as age, sex, or occupation

Why use them?

- To understand who is most affected
- To identify differences hidden in crude rates

Common examples

- Age-specific death rate (ASDR)
- Sex-specific disease rate

Age-Specific Death Rate

Age group	YT population	OT population	YT deaths	OT deaths	YT ASDR	OT ASDR
10-20	20,000	5,000	200	30	1,000	600
20-30	40,000	5,000	150	20	375	400
30-40	15,000	10,000	50	40	333	400
40-50	10,000	10,000	20	20	200	200
50-60	5,000	15,000	15	90	300	600
60-70	5,000	25,000	10	150	200	600
70-80	3,500	20,000	20	250	571	1,250
80-90	1,500	10,000	35	200	2,333	2,000
Crude total	**100,000**	**100,000**	**500**	**800**	**500**	**800**

$$\text{ASDR}_{\text{YT, ages 10-20}} = \frac{200}{20{,}000} \times 100{,}000 = 1{,}000$$

Why Can’t We Stop at Age-Specific Rates?

- Age-specific rates show where differences occur.
- Comparing many age groups across large populations is messy.
- We still want one number to compare populations.
- Standardized or adjusted rates answer: What would the rate be if populations were structured the same way?

Standardized Rates Make Fair Comparisons

What are they?

- Rates adjusted to account for differences in population structure, such as age

Why use them?

- To compare populations fairly
- To remove the effect of different age distributions

How are they created?

- Apply each population’s specific rates to a common standard population.

Standardize Population Structure

1. Choose a standard population age distribution.
2. Use the standard proportions as weights.
3. Apply the weights to each population’s age-specific rates.
4. The result is an adjusted rate, as if the populations had the same age structure.

Age-Adjusted Death Rate

$$\text{Age-adjusted death rate} = \sum(\text{ASDR} \times \text{standard proportion})$$

Age	1940 proportion	1970 proportion	2000 proportion
Under 1 year	0.015343	0.017151	0.013818
1-4 years	0.064718	0.067265	0.055317
5-14 years	0.170355	0.200506	0.145565
15-24 years	0.181677	0.174406	0.138646
25-34 years	0.162066	0.122569	0.135573
35-44 years	0.139237	0.113614	0.162613

Disease Occurrence

Prevalence and Incidence

Measuring Disease Occurrence

Two core measures answer different epidemiologic questions:

- **Prevalence:** How widespread are existing cases?
- **Incidence:** How often do new cases occur?

Both can be crude or specific rates.

Prevalence: The Snapshot

$$P = \frac{C}{N}$$

What is prevalence?

- The proportion with a condition at a point or during a period in time
- Includes existing cases
- Ranges from 0% to 100%

Types of prevalence

- **Point prevalence:** Existing cases at a point in time
- **Period prevalence:** Anyone who was a case at any time during the period

Prevalence Examples

Point prevalence

Prevalence of colds in this class as of today:

- Number of cases: 3
- Class population: 100
- Prevalence: $3/100$
- Percentage: $3/100 \times 100\% = 3\%$

Period prevalence

Eye examination survey, 1979-1981:

- 310 people had cataracts.
- 2,477 people were surveyed.
- Prevalence: $310/2{,}477 = 0.125$
- Percentage: 12.5%

Use of Prevalence Data

What is prevalence useful for?

- Describing current burden
- Planning healthcare services
- Monitoring chronic conditions

Limitation

- Prevalence does not measure the risk of developing disease.

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Incidence: New Cases Over Time

What is incidence?

- Measures new cases only
- Focuses on people at risk
- Captures disease onset

Two forms of incidence

- **Cumulative incidence (CI):** Risk, expressed as a proportion, over a time period
- **Incidence rate (IR):** Rate of occurrence that accounts for person-time

Cumulative Incidence

What it measures: The risk, or proportion, of developing a disease over a specified time period

- **Numerator:** Number of new cases during the time period
- **Denominator:** Population at risk at the start of the period

Key distinction from prevalence

- Cumulative incidence includes new cases only and measures risk.
- Prevalence includes existing cases and measures burden.

Limitation: Cumulative incidence does not adjust for people who enter late or leave early.

Example: Cumulative Incidence

Scenario

- Flu season: Jan-Apr
- 100 elementary students healthy on Jan 1 and at risk
- Same setting: crowded classrooms
- All followed for 4 months
- 10 students get flu

$$CI = \frac{\text{new cases}}{\text{at risk at start}} = \frac{10}{100} = 10\%$$

Interpretation: 10% developed flu during Jan-Apr.

When Follow-Up Time Matters

When CI is not appropriate

- Some people enroll late, some leave early, and absences vary.

Why CI fails here

- Different follow-up time means different time exposed, or time at risk.
- People do not share the same starting point or exposure window.

Use instead

- Use the incidence rate based on person-time when follow-up varies.

Incidence Rate Tracks New Cases Over Time

$$IR = \frac{N}{PT}$$

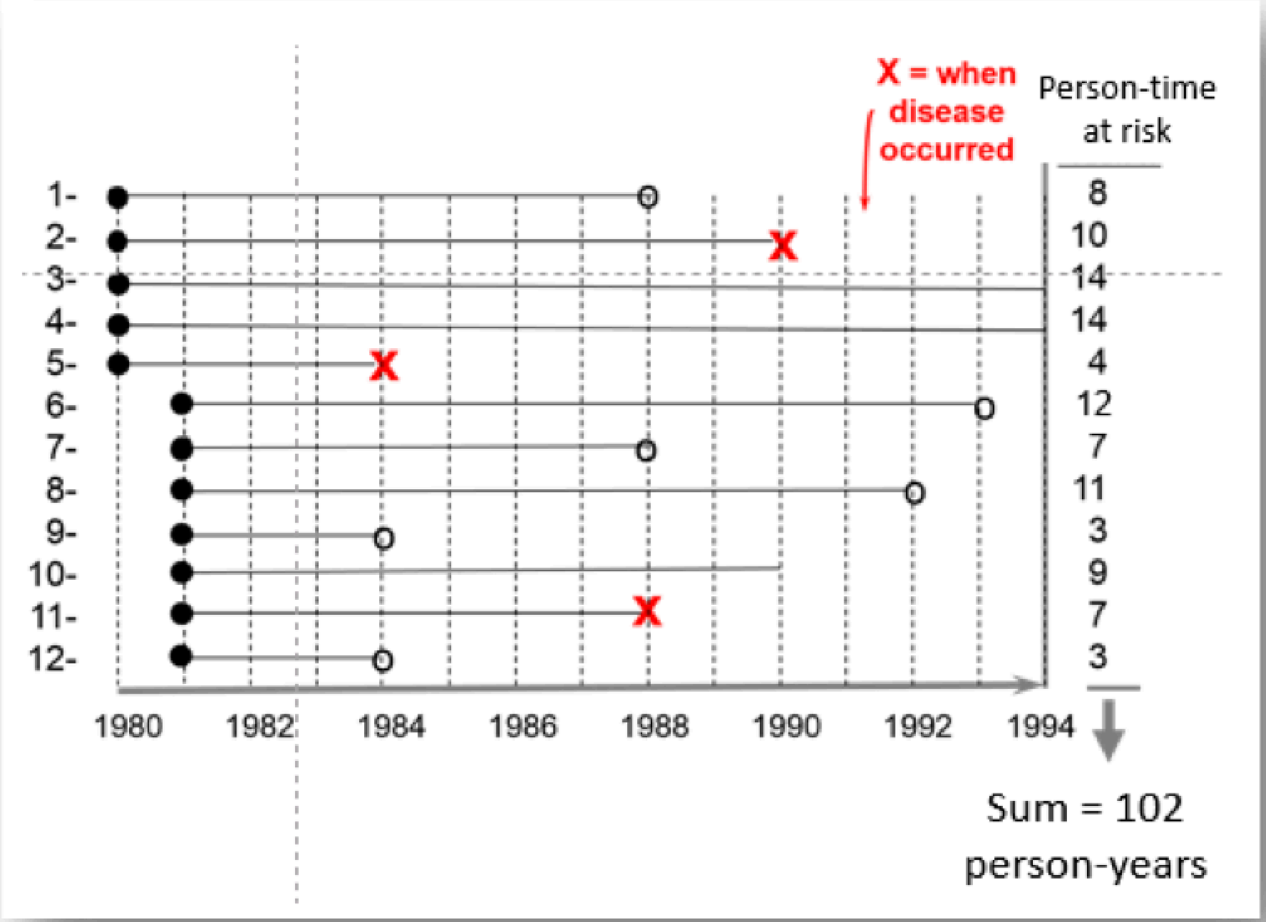
- Measures the speed of new cases
- Uses person-time (PT): the total time each person is at risk and observed
- Example: Person A contributes 1 year, Person B contributes 2 years, and Person C contributes 2 years, for a total of 5 person-years.
- Stop counting person-time when disease occurs, a person is lost to follow-up, or the study ends.

Incidence Rate Example

- New cases: 3
- Total person-time: 102 person-years
- $IR = 3/102 = 0.029$ per person-year
- $IR = 2.9$ per 100 person-years

Interpretation: Over the study period, the disease occurred at a rate of 2.9 new cases per 100 person-years.

In plain language: For every 100 person-years of follow-up, about 3 new cases were observed.



Pain Relief Study: Follow-Up Data

Study setup

- 10 patients receive the new drug and 10 receive the old drug.
- Patients are observed for 10 hours.
- X indicates pain relief; O indicates no relief by hour 10.

Drug	Relief times, hours	No relief by hour 10	Total person-hours
New	1, 1, 1, 1, 2, 3	4	$4(1) + 2 + 3 + 4(10) = 49$
Old	7, 7, 7, 7, 8, 9	4	$4(7) + 8 + 9 + 4(10) = 85$

Pain Relief Study: Questions

1. Which drug provides faster relief? Answer in one sentence.
2. What is the cumulative incidence of relief by 10 hours for each drug?
3. What is the incidence rate of relief per 100 person-hours for each drug?

Pain Relief Study: Answer

Cumulative incidence by 10 hours

- New drug: $6/10 = 60\%$
- Old drug: $6/10 = 60\%$

Incidence rate

- New drug: $6/49 = 12.2$ per 100 person-hours
- Old drug: $6/85 = 7.0$ per 100 person-hours

Conclusion: The new drug is faster because it has a higher incidence rate and less waiting time, even though cumulative incidence is the same. The outcome is beneficial.

Use of Incidence Data

Why incidence matters

- Measures risk
- Tracks outbreaks
- Evaluates interventions
- Complements prevalence

How Are Prevalence and Incidence Related?

Prevalence depends on:

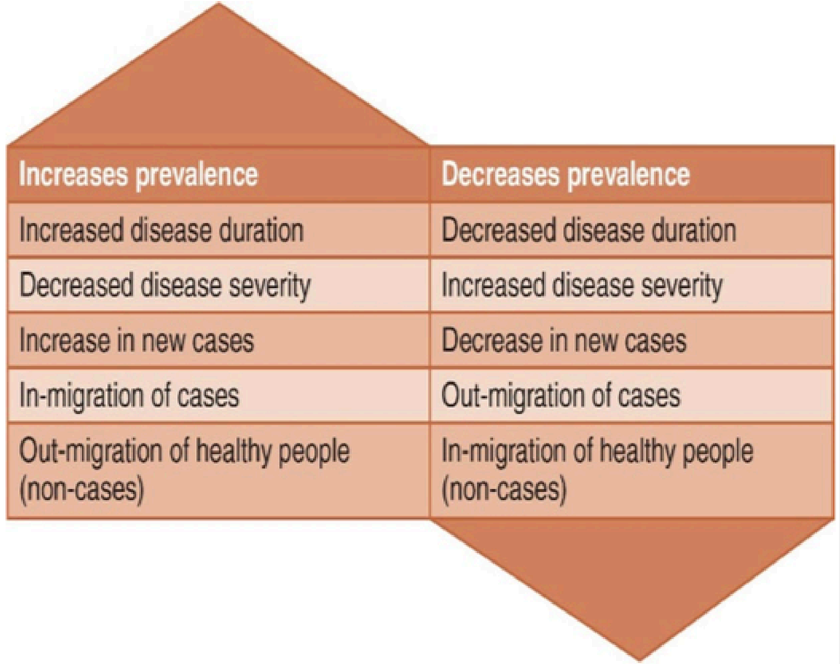
- **Incidence:** How many new cases occur
- **Duration:** How long people remain cases

In a stable population, the rough rule is:

$$P \approx I \times D$$

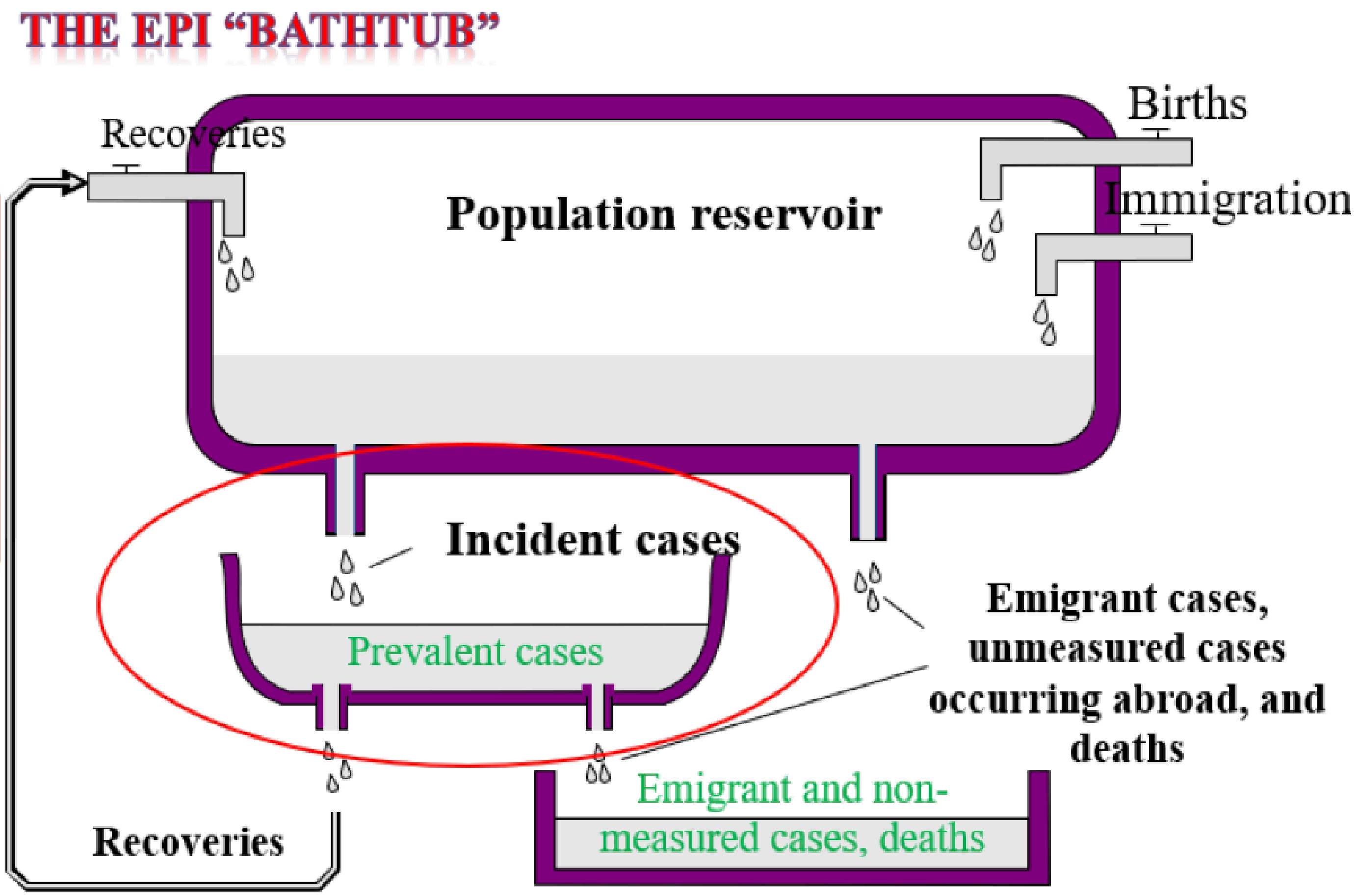
- Higher incidence leads to higher prevalence.
- Longer duration, such as better survival, leads to higher prevalence.

Factors Affecting Prevalence



Increases prevalence	Decreases prevalence
Increased disease duration	Decreased disease duration
Decreased disease severity	Increased disease severity
Increase in new cases	Decrease in new cases
In-migration of cases	Out-migration of cases
Out-migration of healthy people (non-cases)	In-migration of healthy people (non-cases)

Figure 3.8 Factors influencing observed prevalence



Reprinted with permission from Beaglehole R, Kjellström T. Basic Epidemiology. Geneva: World Health Organization; 1993:17.

Source: Beaglehole and Kjellstrom (1993), p. 17.

Mortality

Death Rates

Mortality: Overview

Mortality rate: The frequency of death in a population over time

Mortality rates are used to:

- Compare populations
- Track trends
- Identify leading causes of death

Crude Death Rate: The Simplest Measure

Definition

- Crude death rate (CDR) equals all deaths in a time period divided by the total population.
- Express CDR per 1,000 or 100,000 people.
- Use the mid-year population as the reference population denominator.

$$CDR = \frac{\text{number of deaths in a given year}}{\text{reference population at mid-year}} \times 100{,}000$$

Key point: CDR is a quick summary, but it is affected by age and sex structure.

Sample calculation for the United States in 2020:

$$\frac{3{,}383{,}729}{331{,}449{,}520} \times 100{,}000 = 1{,}020 \text{ per } 100{,}000$$

Specific or Adjusted Death Rate

Specific death rate, such as age- or sex-specific

- Compare within the same subgroup.
- Example: deaths per 100,000 among people age 65 and older

Adjusted or standardized death rate

- Combines age-specific rates using a standard population.
- Compares populations as if they had the same age structure.

Quick discussion: Two cities have the same crude death rate, but one city is older. What measure would you use next - specific or adjusted - and why?

Cause-Specific Mortality Rate

Definition

- Deaths from a specific cause divided by the population
- Used to rank or compare leading causes of death

$$CSMR = \frac{\text{deaths from cause X}}{\text{mid-year population}} \times 100{,}000$$

Sample calculation: Disease X in City A, 2023

- Deaths from Disease X: 150
- Mid-year population: 500,000
- $CSMR = 150/500{,}000 \times 100{,}000 = 30$ deaths per 100,000

Case-Fatality Rate

Definition

- Deaths among diagnosed cases divided by total cases during a period
- Used to measure severity: the probability of death among cases

$$CFR = \frac{\text{deaths among cases}}{\text{total cases}} \times 100\%$$

Sample calculation: Disease X in City A, 2023

- Diagnosed cases: 1,500
- Deaths among cases: 150
- $CFR = 150/1{,}500 \times 100\% = 10\%$

Proportional Mortality Ratio

Definition

- Deaths from cause X divided by all deaths in the same period
- Answers: How important is this cause among all deaths?
- PMR is not a risk rate because it has no population denominator.

$$PMR = \frac{\text{deaths from cause X}}{\text{all deaths}} \times 100\%$$

Sample calculation for the United States:

- Heart disease deaths in 2013: 611,105
- All-cause deaths in 2013: 2,596,993
- $PMR = 611{,}105/2{,}596{,}993 \times 100\% = 23.5\%$

Mortality Comparison Table

Measure	Numerator	Denominator	Best for	Common mistake
CDR	All deaths	Total population	Overall comparison	Ignoring age structure
CSMR	Deaths from cause X	Total population	Leading causes	Confusing with PMR
CFR	Deaths among cases	Total cases	Severity	Using a population denominator
PMR%	Deaths from cause X	All deaths	Cause contribution	Treating as risk

Overview of Mortality Measures

General population

- Annual death rate
- Crude death rates
- Cause-specific death rates
- Adjusted death rates
- Case-fatality rate
- Standardized mortality ratio
- Proportional mortality ratio (PMR%)
- Mortality crossover and mortality time trends

Maternal and infant population

- Maternal mortality rate
- Infant mortality rate
- Fetal mortality
  - Fetal death rate
  - Late fetal death rate
- Perinatal mortality rate
- Birth rate, such as crude birth rate
- General fertility rate

Key Takeaways: Counts and Rates

- **Counts** tell us how many events occurred.
- **Rates** describe events relative to a population and often time.
- A **crude rate** is an overall snapshot.
- A **specific rate** describes a subgroup, such as an age or sex group.
- An **adjusted rate** supports fair comparisons across different population structures.

Key Takeaways: Prevalence and Incidence

- **Prevalence** measures existing cases and burden.
- **Incidence** measures new cases and risk.
- **Cumulative incidence:** What proportion became cases?
- **Incidence rate:** How fast are new cases occurring? It uses person-time.
- In a stable setting, prevalence depends on incidence and duration: $P \approx I \times D$.

Key Takeaways: Mortality

- **CDR:** Overall deaths in a population; affected by age and sex structure
- **CSMR:** Deaths from a specific cause in the population
- **CFR:** Deaths among cases; a measure of severity
- **PMR%:** Share of deaths due to a cause; not a measure of risk
- Use specific or adjusted death rates when populations differ in structure.

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References

Beaglehole, Robert, and Tord Kjellstrom. 1993. *Basic Epidemiology*. World Health Organization.

Tokis, Sandra L. 1999. *Titanic: A Statistical Exploration*.